Section A

Multiple Choice Questions **(8 X 0.5 = 4 Ma**

1. Which one of the following is the Information security goal?

   A. Identification
   B. Prevention
   C. Security
   D. Prediction

2. "A person looking over our shoulder while we type a password" – Which one of the Information security concept stated by CIA trad is compromised here?

   A. Authenticity
   B. Accuracy
   C. Confidentiality
   D. Integrity

3. Identify the first step that must be done in a Biometric Authentication Process?

   A. Extraction of features
   B. Acquiring the sample
   C. Identify the Object
   D. Matching with the existing sample

4. Which one of the following uses multifactor authentication.

   A. Login to a session using password
   B. Opening a door with facial recognition
   C. Sending OTP for the Bank Transaction
   D. Using an access card to enter the office

5. Identify the access that follows Least Privilege

   A. Students are allowed to access all applications in the college website
   B. eLearning access is given to the students for the courses they are registered only
   C. Admin staff are given the privilege to enter marks of the students
   D. Student login accounts have capabilities as the admin

6. "When the software with access to a resource has a greater level of permission to access the resource than the user who is controlling the software" – Which one of the following attacks is referred by the given statement?

   A. Spoofing
   B. Eavesdropping
   C. Confused Deputy Problem
   D. Sniffing Problem

7. Which one of the following performs strictly as a monitoring and alert tool?

   A. Antivirus Program
   B. Operating System
   C. Intrusion Detection System
   D. Virtual Private Network

8. Identify the statement that refers to Accountability

   A. Mechanism to allow or deny a user
   B. Way of verifying the Identity of a user
   C. A model used for authorizing a user
   D. Helps to know who a given transaction is associated with

Section B

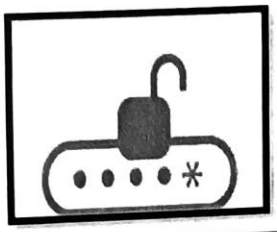
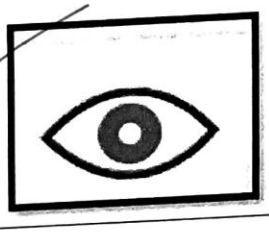
Short Answer Questions **(4* 2 = 8 marks)**

1. For the given scenario, identify the breach of the CIA Triad (confidentiality, Integrity and Availability) [0.5 Marks Each]

S. No	Scenario	Breach of the CIA Triad
A.	A hospital under the ministry of health allows the sensitive data of a patient to be shared among all the staff.	Confidentiality
B.	An employee in the company deleted the records of another employee using unauthorized means of access.	Integrity
C.	AVM cloud services systems have a delay in the network due to the increased number of users who login into the system.	Availability
D.	Students in the university are allowed to access transcripts of other students.	Confidentiality

2

2. Refer to the images given below and find out the authentication factor used. [0.5 Marks Each].

Question No	Authentication Image	Authentication factor
A.		Something you have
B.		Something you Know
C.		where you are
D.		Something you are

[1 Mark Each]

3. List out any two Access control Models.

Manual access control (MAC)

Atribute based access control (ABAC)

[1 Mark Each]

4. Write down two reasons for performing Auditing.

- to ensure that the employees are doing their work
- to help keeping the Network safe
- detect any unuthrized access

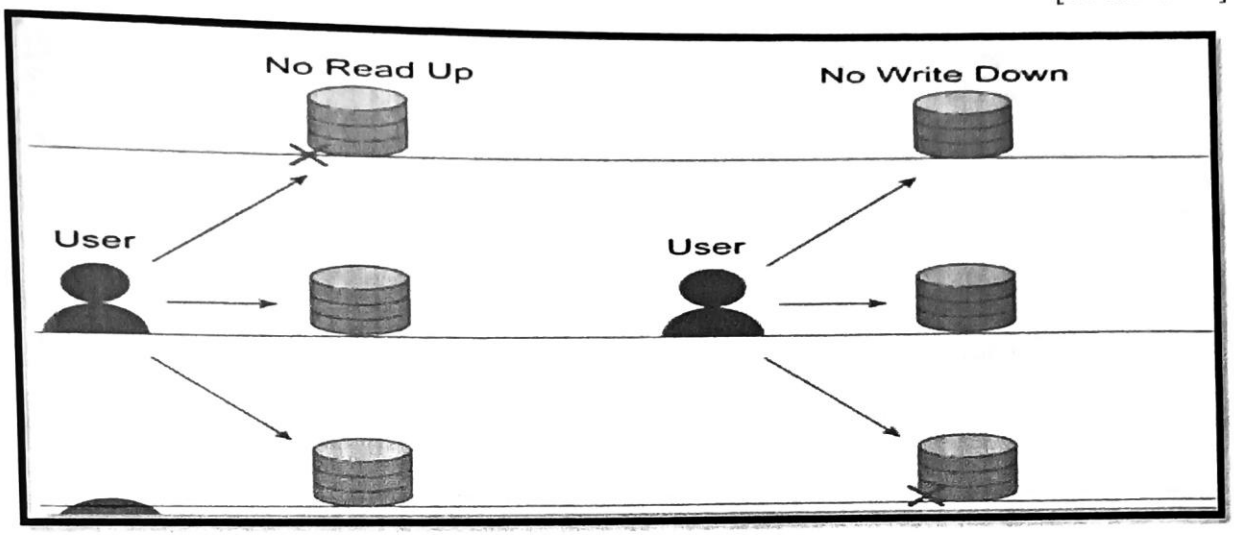
Scenario Based Questions

Section C

(2* 4 = 8 marks)

1.a) The following diagram illustrates the access given to the users of AYZ LLC. on the critical data. Answer the questions given below.

[1 Mark Each]



A. Which Multilevel Access control model implementation is shown in the above diagram?

Bill - lapdula model

B. What are the two properties of the above model illustrated in the diagram (List out)?

- the simple security property
- the * property

1.b) Refer to the given diagram and answer the questions below.

```
[root]# touch sub_file
[root]# ls -l sub_file
-rw-r--r--+ 1 root 0 Jul  9 03:06 sub_file
[root]# getfacl --omit-header sub_file
user::rw-
group::r-x                          #effective:r--
group:stu_gp:r--
mask::r--
other::r--
```

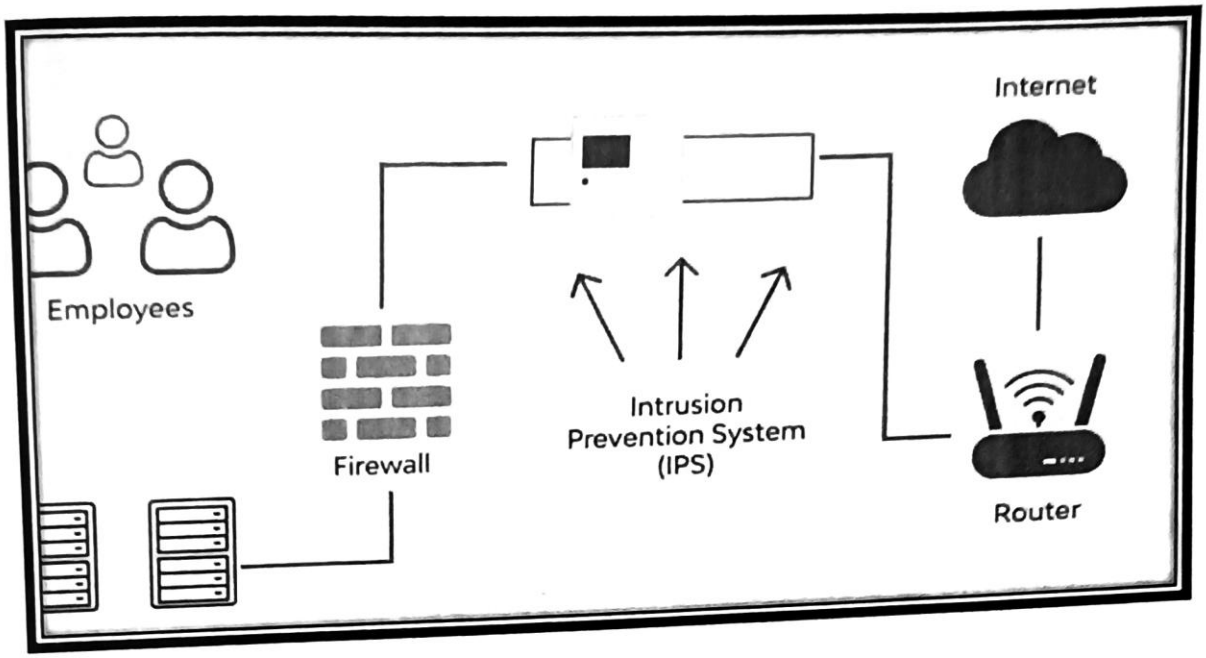
A. Which ACL (Access Control List) is used here?

Unix Linux File Acl

B. What are the permissions specified here?

Read, write, exclude

2.) The following diagram shows the network topology of the company ARENA MNC that implements thorough accountability and auditing.



A. Which benefit of accountability helps us to prevent an employee denying his actions? [0.5 Mark]

denying access deterrence

B. Identify any two possible locations where the log can be collected. [1 Mark]

- Network
- System

C. Which tool in this topology must be used to take actions based on what is happening in the environment? [0.5 Mark]

Intrusion prevention system

D. Who gets access to the logs that are recorded in the network? [0.5 Mark]

adminstrators

E. Is it possible to change the logs that are recorded? [0.5 Mark]

NO

F. Identify any two critical events where logging mechanism can be configured? [1 Mark]

- Hardware errors
- Software failure
- login and logout

3½